

RAJESH K. JAT

AI Engineer

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Professional Summary

AI Engineer specializing in Generative AI, LLM-based applications, and multimodal intelligence systems. Experienced in building self-hosted RAG pipelines, LangChain-based orchestration, and CUDA-optimized vLLM deployments in air-gapped environments. Skilled in full-stack GenAI systems integrating TTS, vision, and document intelligence.

Core Competencies

- **LLM & RAG Systems:** LangChain, LlamaIndex, vLLM, LLaMA, Mistral, CrewAI, H2OGPT, HuggingFace Transformers.
- **Multimodal AI:** Text-Vision-Speech integration, Coqui TTS, OCR, Vision-Language Models (VLMs).
- **Backend/API Development:** FastAPI, REST APIs, Dockerized microservices, Linux server administration.
- **Edge AI Deployment:** NVIDIA CUDA, on-premise & air-gapped setups, low-latency optimization.
- **Data Engineering:** Vector databases (Milvus, ChromaDB, FAISS), structured/unstructured document retrieval.

Experience

AI Research Engineer

Jul 2023 – Present

WESEE – Naval R&D Lab, New Delhi

- Led the design of a self-hosted **GenAI Web App** using LLaMA with LangChain RAG pipeline over a classified naval corpus.
- Built FastAPI-based backend and ReactJS frontend, containerized via Docker, deployed in **air-gapped, CUDA-enabled Linux servers**.
- Integrated vLLM for optimized inference, achieving 40% lower latency and memory footprint.
- Architected document-to-audio conversion pipeline using **Coqui TTS + LLaMA**, generating podcast-style summaries.
- Developed a multimodal **PDF intelligence assistant** enabling Q&A, summaries, and FAQs from uploaded documents.

AI Intern

Sep 2022 – Jan 2023

Object Automation System Solutions Pvt. Ltd., Chennai

- Built a Flask-based recommendation engine using collaborative filtering.
- Hosted and administered a Linux VPS for LMS deployment, ensuring 24/7 reliability.

Projects

Multimodal PDF Assistant

Created a RAG-powered document assistant with LLaMA 3.1 and LangChain. Integrated Coqui TTS for natural speech generation and ReactJS frontend.

Sponsor Analytics for Broadcasts

Developed YOLO-based video analytics to detect brand logos and estimate screen-time for sponsor

ROI metrics.

Education

M.Tech in Artificial Intelligence & Data Science 2021 – 2023

IIIT Kota, MNIT Jaipur Campus

B.Tech in Computer Science & Engineering 2017 – 2021

Rajasthan Technical University

Certifications

- AWS Fundamentals – Coursera
- Developing Cloud Native Applications – Coursera

Technical Skills

- **Languages:** Python, C++, JavaScript, SQL
- **Libraries/Frameworks:** LangChain, LlamaIndex, vLLM, PyTorch, TensorFlow, FastAPI, Streamlit, ReactJS
- **DevOps/Tools:** Docker, Git, GitLab (Self-hosted), NVIDIA CUDA, Linux Administration
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Platforms:** Ubuntu, AWS